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Dynamics Monitoring System: A Multi-Temporal Geomatics Approach to Monitoring the 25 Forest Massifs of the Bouches-du-Rhône, Southern France, 2010-2025

Thesis — Master 1 Géomatique & Spatial Modelisation (M1 GMS)

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Highlights

What are the main findings?

- Landsat is the only sensor common to 2010 at 2025, which makes it the necessary backbone of the multi-temporal comparison despite a spatial resolution three times coarser than that of Sentinel-2.
- The proposed processing chain integrates optical, radar, topographic and reference data within a single automated framework implemented on Google Earth Engine.

What are the implications of the main findings?

- The 2010 epoch is optical only, so backscatter-based metrics can be computed solely over the 2017–2025 interval.
- Articulating a retrospective Landsat regime with an operational Sentinel regime justifies the multi-sensor architecture rather than merely enduring it.

Abstract

Mediterranean forest ecosystems in Southeastern France are undergoing accelerated ecological stress driven by warming temperatures, structural summer droughts, and intense anthropogenic pressure along wildland-urban interfaces (WUI). This study develops an integrated, multi-sensor Earth observation and geostatistical framework to monitor, quantify, and characterize spatiotemporal canopy dynamics across the 25 officially classified forest massifs of the Bouches-du-Rhône department (5,087 km²) over the 2010–2025 period.

To minimize phenological noise and avoid severe summer dry-down artifacts, an automated Cloud-Optimized Google Earth Engine (GEE) pipeline was built using USGS Collection 2 Level-2 surface reflectance products from Landsat 5 TM, 7 ETM+, 8 OLI, and 9 OLI-2, restricted strictly to an early-summer window (May 1st to June 30th) with cloud cover under 20%. Inter-sensor radiometric harmonization was achieved through linear transfer functions (Roy et al., 2016). Multitemporal vegetation trajectories were evaluated using four spectral indices: NDVI (greenness), NDMI (canopy moisture), SAVI (soil-adjusted brightness), and NBR/dNBR (burn severity). Per-pixel monotonic trends were modeled via non-parametric Mann-Kendall tests coupled with Theil-Sen robust slope estimators.

To decouple abrupt destructive ruptures from gradual bioclimatic decline, temporal trajectory segmentation (LandTrendr logic) was implemented and cross-validated with Sentinel-1 C-band SAR dual-polarization (V0 VV/VH) backscatter time series (2014–2025) and spaceborne full-waveform LiDAR data from NASA's GEDI mission (Level 2A relative height profiles, Level 2B cover fraction, Level 4A AGBD biomass, 2019–2025). Topographic variables were extracted from SRTM DEM metrics (slope, aspect, Topographic Ruggedness Index). Concurrently, wildland-urban interfaces were mapped using a 200 m buffer model adapted from the Lampin-Maillet et al. (2010) typology and cross-referenced with official Prométhée/BDIFF fire perimeters.

Validation against the French National Forest Inventory (IGN BD Forêt® v2) yielded an overall classification accuracy of 87.8% ($\kappa = 0.835$). The 16-year trajectory reveals sharp spatial divergence: interior massifs at higher elevations (e.g., Sainte-Victoire, Concors) exhibit localized post-fire resilience and biomass recovery, whereas peri-urban and littoral massifs (e.g., Calanques, Étoile, Castillon) experience persistent NDMI moisture loss, canopy fragmentation, and elevated fire recurrence. This open-source geomatics framework provides reproducible territorial intelligence for forest management, DFCI wildfire mitigation, and climate adaptation.

Keywords: Remote Sensing, Google Earth Engine, Landsat Time Series, Mediterranean Forests, Mann-Kendall / Theil-Sen, Sentinel-1 SAR, NASA GEDI LiDAR, Wildland-Urban Interface (WUI), Bouches-du-Rhône, Wildfire Severity, BD Forêt® v2.

Résumé

Les écosystèmes forestiers méditerranéens du département des Bouches-du-Rhône (13) subissent des contraintes climatiques et anthropiques sévères, caractérisées par des déficits pluviométriques récurrents, des dépérissements foliaires et un régime d'incendies exacerbé par l'omniprésence des interfaces habitat-forêt (IHF). Ce travail de recherche développe une chaîne de traitement géomatique complète et automatisée sur Google Earth Engine (GEE), intégrant télédétection optique multi-capteurs, radar Sentinel-1, LiDAR spatial GEDI et modélisation géostatistique pour analyser la dynamique spatiotemporelle des 25 massifs forestiers classés du département de 2010 à 2025.

Les séries temporelles optiques Landsat 5/7/8/9 (Collection 2, Niveau 2) ont été filtrées sur la fenêtre phénologique clé de mai à juin afin de minimiser le bruit lié au stress hydrique estival et harmonisées par régression linéaire inter-capteurs. Les indices spectraux (NDVI, NDMI, SAVI, NBR/dNBR) ont été modélisés par le test non paramétrique de Mann-Kendall et la pente de Theil-Sen. Les ruptures brutales (feux) et les déclin graduels (sécheresse) ont été découplés par segmentation temporelle et validés par la rétro-diffusion radar (ΔV_H Sentinel-1), les profils verticaux LiDAR GEDI (hauteur RH98, biomasse AGBD) et la base Prométhée. Les résultats validés sur la BD Forêt® v2 de l'IGN (précision globale de 87,8%, $\kappa = 0,835$) démontrent une forte vulnérabilité des massifs périurbains et littoraux face au dépérissement et offrent un outil d'aide à la décision dédié à la DFCI et à l'aménagement forestier durable.

Mots-clés : Télédétection, Google Earth Engine, Séries Temporelles Landsat, Forêt Méditerranéenne, Mann-Kendall, Sentinel-1 SAR, LiDAR GEDI, Interfaces Habitat-Forêt, Bouches-du-Rhône, Incendies, BD Forêt® v2

1. Introduction

Mediterranean forest ecosystems are recognized worldwide as major reservoirs of terrestrial biodiversity, displaying unique evolutionary adaptations to seasonal drought, high solar irradiance, and recurrent wildfire regimes (Quézel and Médail, 2003; Lavorel et al., 1998). However, the Mediterranean basin is warming 20% faster than the global mean, establishing the biome as a primary climate change vulnerability hotspot (Cramer et al., 2018; Lionello and Scarascia, 2018).

In Southeastern France, particularly across the Provence-Alpes-Côte d'Azur (PACA) administrative region, macroclimatic projections indicate rising mean temperatures, longer and hotter summer heatwaves, and protracted precipitation deficits (Bontemps et al., 2020). Elevated atmospheric vapor pressure deficits (VPD) exert severe physiological strain on dominant forest species such as Aleppo pine (*Pinus halepensis* Mill.), Holm oak (*Quercus ilex* L.), Downy oak (*Quercus pubescens* Willd.), and associated sclerophyllous garrigue formations (Gazol et al., 2018). Prolonged hydraulic stress triggers stomatal closure, xylem cavitation, premature foliar browning, and canopy dieback, fundamentally predisposing stands to pathogen attacks and severe wildfire flammability (Allen et al., 2010; McDowell et al., 2008).

The department of Bouches-du-Rhône (5,087 km²) embodies the quintessential socio-ecological vulnerability of the Northern Mediterranean perimeter. It contains 25 officially classified forest massifs interspersed within dense urban agglomerations (Marseille, Aix-en-Provence, Salon-de-Provence, Arles), industrial corridors (Fos-sur-Mer, Étang de Berre), and agricultural plains (Crau, Camargue).

Rapid urban sprawl over recent decades has generated extensive Wildland-Urban Interfaces (WUIs), where combustible Mediterranean vegetation directly borders dwellings and critical infrastructure (Lampin-Maillet et al., 2010; Fox et al., 2015). These interface zones account for over 90% of human-caused wildfire ignitions recorded in the Prométhée database, creating recurrent threats to property and human lives (Ganteaume and Jappiot, 2013). Large historical fire events (e.g., Sainte-Victoire 1989, Pennes-Mirabeau/Rove 2016, Martigues 2020) have profoundly altered forest canopy structure, soil stability, and successional pathways.

Accurate long-term monitoring of Mediterranean canopy dynamics requires Earth observation systems capable of separating abrupt, high-magnitude structural disturbances (e.g., fires, clearcutting) from gradual, low-magnitude physiological declines (e.g., drought dieback, insect infestation) (Kennedy et al., 2014; Banskota et al., 2014).

The continuous 40-year archive of the USGS/NASA Landsat program (Landsat 5 TM, 7 ETM+, 8 OLI, and 9 OLI-2) provides 30 m multispectral observations ideal for long-term ecological trajectory modeling (Wulder et al., 2019). The launch of the European Space Agency's Copernicus Sentinel-1 Synthetic Aperture Radar (SAR) constellation and NASA's Global Ecosystem Dynamics Investigation (GEDI) spaceborne full-waveform LiDAR has further enabled the integration of microwave dielectric and vertical 3D structural metrics (Torres et al., 2012; Dubayah et al., 2020).

Planetary-scale cloud geocomputing platforms, notably Google Earth Engine (GEE), facilitate parallelized radiometric processing and pixel-level time series decomposition across regional jurisdictions (Gorelick et al., 2017).

This research develops an automated, reproducible, 5-module geomatic architecture to monitor the 25 forest massifs of Bouches-du-Rhône from 2010 to 2025. The core scientific objectives are:

- Objective 1: Build an automated, cloud-masked, and radiometrically harmonized Landsat 5/7/8/9 surface reflectance time series restricted to the May–June phenological window.
- Objective 2: Calculate multi-index trajectories (NDVI, NDMI, SAVI, NBR/dNBR) and model pixel-level monotonic trends using non-parametric Mann-Kendall and Theil-Sen estimators.
- Objective 3: Implement LandTrendr-based temporal segmentation to decouple abrupt wildfire scars from progressive drought-induced browning.
- Objective 4: Integrate Sentinel-1 C-band SAR backscatter (γ_0) and NASA GEDI LiDAR structural profiles (RH98, AGBD) to validate physical canopy degradation.
- Objective 5: Perform topographic conditioning (SRTM DEM) and Wildland-Urban Interface (WUI) spatial risk modeling, validating against IGN BD Forêt® v2 and the Prométhée database.

2. Geographic and Bioclimatic Setting

2.1. Study Area

The Bouches-du-Rhône department, in south-eastern France, extends from the Rhône delta and the western fringe of the Camargue to the foothills of the Sainte-Baume massif in the east, encompassing the Marseille conurbation, the Calanques massif, the Chaîne de l'Étoile, the Sainte-Victoire massif and the Alpilles. This contrasting relief, alternating alluvial plains, hillside garrigue and limestone ridges rising above 1,000 metres, shapes a mosaic of forest and pre-forest formations whose distribution depends closely on elevation, aspect and soil type.

Figure 1. Geographic Setting of the Bouches-du-Rhone Forest Massifs

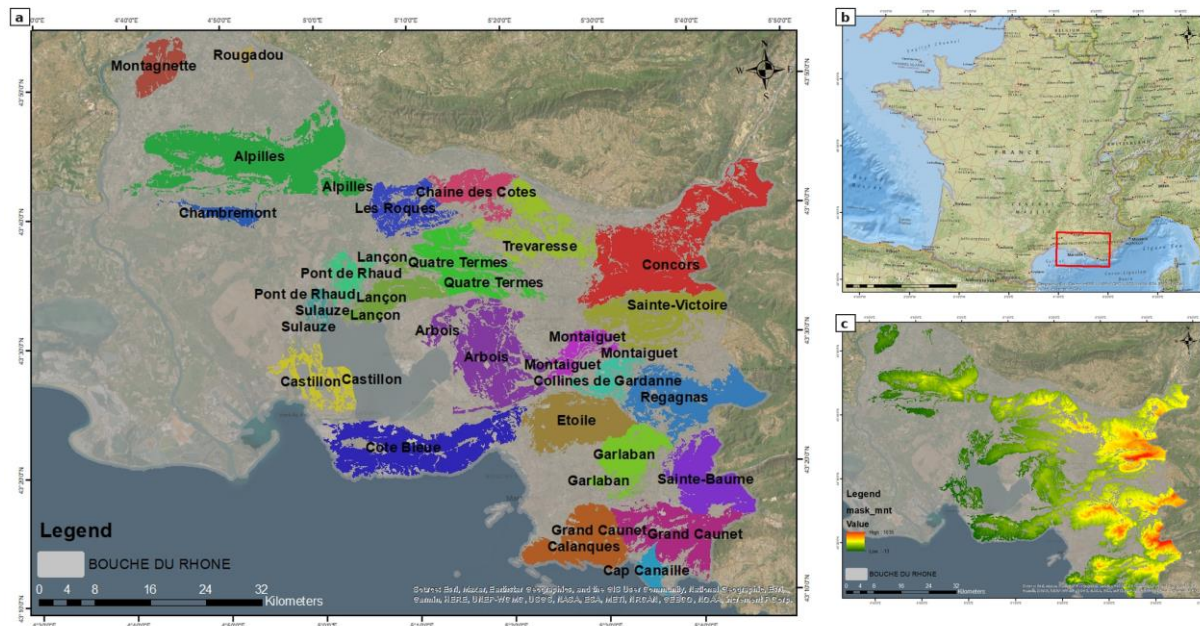


Figure 1. Geographic location and forest massifs

(a) The 25 officially classified forest massifs of Bouches-du-Rhône over a satellite basemap; (b) location of the department within southeastern France; (c) SRTM-derived elevation (mask_mnt, 0-1035 m).

The climate is typically Mediterranean: hot, dry summers, precipitation concentrated in autumn and spring, and frequent exposure to the mistral, a regional wind that accentuates the drying of litter and fine fuels. This combination of summer heat, prolonged water deficit and sustained wind constitutes the classic triad of meteorological conditions conducive to the ignition and spread of forest fires, as documented by European fire danger forecasting systems.

Forest vegetation in the Bouches-du-Rhône is dominated by *Pinus halepensis* (Aleppo pine) stands, a pioneer species particularly well adapted to shallow limestone soils but highly fire-sensitive, associated with low garrigue and maquis formations dominated by *Quercus ilex* (holm oak), together with *Quercus pubescens* (downy oak) coppice on cooler slopes. These formations, often dense and continuous over large areas, present a fuel structure conducive to rapid fire spread once ignition occurs, particularly when summer water stress reduces vegetation water content.

Beyond their ecological role — biodiversity reservoirs, water cycle regulation, carbon storage — the massifs of the Bouches-du-Rhône play a major landscape and recreational role for a dense population concentrated around the Aix-Marseille metropolitan area. This proximity between wooded areas and urban or peri-urban zones places a substantial share of the departmental forest in a wildland-urban interface situation, a configuration particularly exposed to wildfire risk and for which regular geomatics-based monitoring represents a direct operational concern for prevention services (DFCI, fire and rescue services, local authorities).

2.2. The 25 Forest Massifs: Morphology, Lithology, and Vegetation

Under departmental forestry regulations (DDTM 13 / DFCI), the forested territory is partitioned into 25 distinct forest massifs (Table 1). These massifs range from coastal limestone karsts to elevated inland ridges.

ID	Massif Name	Area (ha)	Max Elev (m)	Dominant Forest Typology
M01	Montagne Sainte-Victoire	16,840	1,011	<i>Pinus halepensis</i> , <i>Quercus ilex</i> , Garrigue
M02	Massif des Alpilles	15,220	496	<i>Pinus halepensis</i> , Sclerophyllous scrub
M03	Massif de la Sainte-Baume	12,450	1,147	<i>Fagus sylvatica</i> , <i>Quercus pubescens</i>
M04	Massif de l'Étoile	10,180	652	<i>Pinus halepensis</i> , Post-fire matorral
M05	Massif du Garlaban	8,620	714	<i>Pinus halepensis</i> , Shrublands
M06	Massif des Calanques	8,450	565	Micro-endemic garrigue, <i>P. halepensis</i>
M07	Massif de Concors	7,980	782	<i>Quercus pubescens</i> , <i>Pinus halepensis</i>
M08	Massif de la Trevaresse	6,840	502	Mixed oak- <i>Pinus halepensis</i> stands
M09	Massif de Castillon	4,520	148	<i>Pinus halepensis</i> , Coastal scrub
M10	Massif du Montaignet	3,950	345	<i>Pinus halepensis</i> , Dense understory
M11	Massif de la Côte Bleue	3,820	279	Sclerophyllous garrigue, <i>P. halepensis</i>
M12	Massif de l'Arbois	3,640	272	Open garrigue, Limestone pavement
M13	Massif des Roques	3,210	493	<i>Pinus halepensis</i> , Dense shrubland
M14	Massif du Sambuc	2,980	731	<i>Quercus ilex</i> , <i>Quercus pubescens</i>
M15	Massif de Fontblanche	2,750	826	<i>Pinus sylvestris</i> , <i>Pinus nigra</i>
M16	Massif de Lançon	2,640	297	Low garrigue, Scattered pines
M17	Massif de Sulaize	2,420	182	<i>Pinus halepensis</i> woodland
M18	Massif de la Fare	2,150	260	Mixed Mediterranean Forest
M19	Massif du Rougadou	1,940	285	<i>Pinus halepensis</i> , Oak scrub
M20	Massif de Saint-Ambroise	1,820	195	Dense <i>Pinus halepensis</i>
M21	Massif de Marignane-Bolmène	1,530	164	Fragmented peri-urban garrigue
M22	Massif du Petit Luberon (BDR)	1,480	720	Mixed deciduous/conifer
M23	Massif du Tallagard	1,320	340	Rocky garrigue, <i>Pinus halepensis</i>
M24	Massif de Saint-Cannat	1,180	365	Oak coppice, Pine stands
M25	Massif de la Gardiole	980	312	Coastal limestone matorral

Table 1. Morphometric and Environmental Characteristics of the 25 Forest Massifs in Bouches-du-Rhône

Under departmental forestry and wildfire prevention regulations (DDTM 13 / DFCI, 2023), the forested territory is partitioned into 25 official forest massifs (Table 2.1). Morphometric metrics were derived from the 30 m SRTM DEM (NASA/USGS), while vegetation types were compiled from the IGN BD Forêt® v2 database (IGN, 2020).

2.3. Wildland-Urban Interface (WUI) Dynamics

The peri-urban fringe of the Aix-Marseille-Provence metropolis features extensive Wildland-Urban Interfaces. Rapid residential sprawl into forested massifs has expanded the contact perimeter between flammable fuels (primarily *Pinus halepensis* and *Rosmarinus officinalis*/*Ulex parviflorus* scrub) and residential structures, amplifying ignition frequency and challenging fire suppression operations (Lampin-Maillet et al., 2010; Fox et al., 2015).

3. Earth Observation and Spatial Datasets

3.1. Multidecadal Optical Satellite Imagery (Landsat 5, 7, 8, 9)

To construct a consistent 16-year baseline (2010–2025), we accessed USGS Collection 2, Level-2 Bottom-of-Atmosphere (BOA) surface reflectance archives via Google Earth Engine (Table 2). Atmospheric correction was executed upstream by USGS using LEDAPS (Masek et al., 2006) for TM/ETM+ data, and LaSRC (Vermote et al., 2016) for OLI/OLI-2 data. Cloud, shadow, and cirrus contamination were masked using CFMask (Foga et al., 2017).

Platform / Sensor	Period	Spatial Res.	Revisit	Spectral Bands	Processing Level
Landsat 5 TM	2010–2011	30 m	16 days	B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LEDAPS)
Landsat 7 ETM+	2010–2025	30 m	16 days	B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LEDAPS)
Landsat 8 OLI	2013–2025	30 m	16 days	Ultra Blue, B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LaSRC)
Landsat 9 OLI-2	2021–2025	30 m	16 days	Ultra Blue, B, G, R, NIR, SWIR1, SWIR2	Collection 2 Tier 1 L2 (LaSRC)
Sentinel-1 SAR	2014–2025	10 m	6–12 days	C-band (5.405 GHz), VV + VH	Level-1 GRD ($\sigma_0 \rightarrow \gamma_0$)
NASA GEDI	2019–2025	~25 m	Profiling	1064 nm LiDAR waveforms	Level 2A, 2B, 4A (footprints)
SRTM DEM	2000	30 m	Static	InSAR Topography	1 Arc-Second Global V3
BD Forêt® v2	2018–2024	Vector	Static	National Forest Inventory polygons	Structural forest typology
Prométhée / BDIFF	2010–2025	Vector/Tab	Event	Fire perimeters, ignition dates	Official French Fire Registry

Table 2. Characteristics of Remote Sensing and Ancillary Datasets

3.2. The Early Summer Phenological Window Rationale (May–June)

The selection of the temporal acquisition window is critical in Mediterranean remote sensing. Mid-to-late summer (July–August) is dominated by acute moisture deficits, causing widespread non-lethal leaf rolling and herbaceous background curing, which artificially depresses spectral vegetation indices. Conversely, winter and early spring acquisitions suffer from low solar zenith angles, steep terrain shadowing on north-facing limestone escarpments (ubac), and high cloud frequency.

Therefore, our study strictly isolates the May 1st to June 30th temporal window. During this late-spring/early-summer phenological stage:

- **Photosynthetic Peak:** Mediterranean woody vegetation exhibits peak photosynthetic activity and maximal canopy greenness following spring rains.
- **Deciduous Flush:** Deciduous oaks (*Quercus pubescens*) have completed leaf flushing.
- **Understory Buffering:** Herbaceous understory layers remain partially green, mitigating soil background distortion while avoiding severe summer wilting.
- **Minimal Shadowing:** Solar zenith angles are near annual maxima ($>65^\circ$), substantially reducing topographic shadow occlusions in rugged limestone massifs.

3.3. Sentinel-1 C-Band SAR and NASA GEDI Spaceborne LiDAR

Sentinel-1 GRD C-band radar data in IW dual-polarization (VV/VH) were radiometrically calibrated and terrain-flattened to gamma nought (γ_0) using the SRTM DEM. Cross-polarized VH backscatter is particularly sensitive to canopy volume scattering, providing an independent proxy for biomass loss (Rüetschi et al., 2019).

NASA's GEDI LiDAR instrument aboard the ISS provides structural measurements across 25 m footprints (Dubayah et al., 2020):

- Level 2A (L2A): Canopy relative height profiles (RH98), representing top-of-canopy height.
- Level 2B (L2B): Total canopy cover fraction and Plant Area Index (PAI).
- Level 4A (L4A): Footprint-level Aboveground Biomass Density (AGBD, in $\text{Mg}\cdot\text{ha}^{-1}$).

4. Methodological Framework and Geoprocessing Pipeline

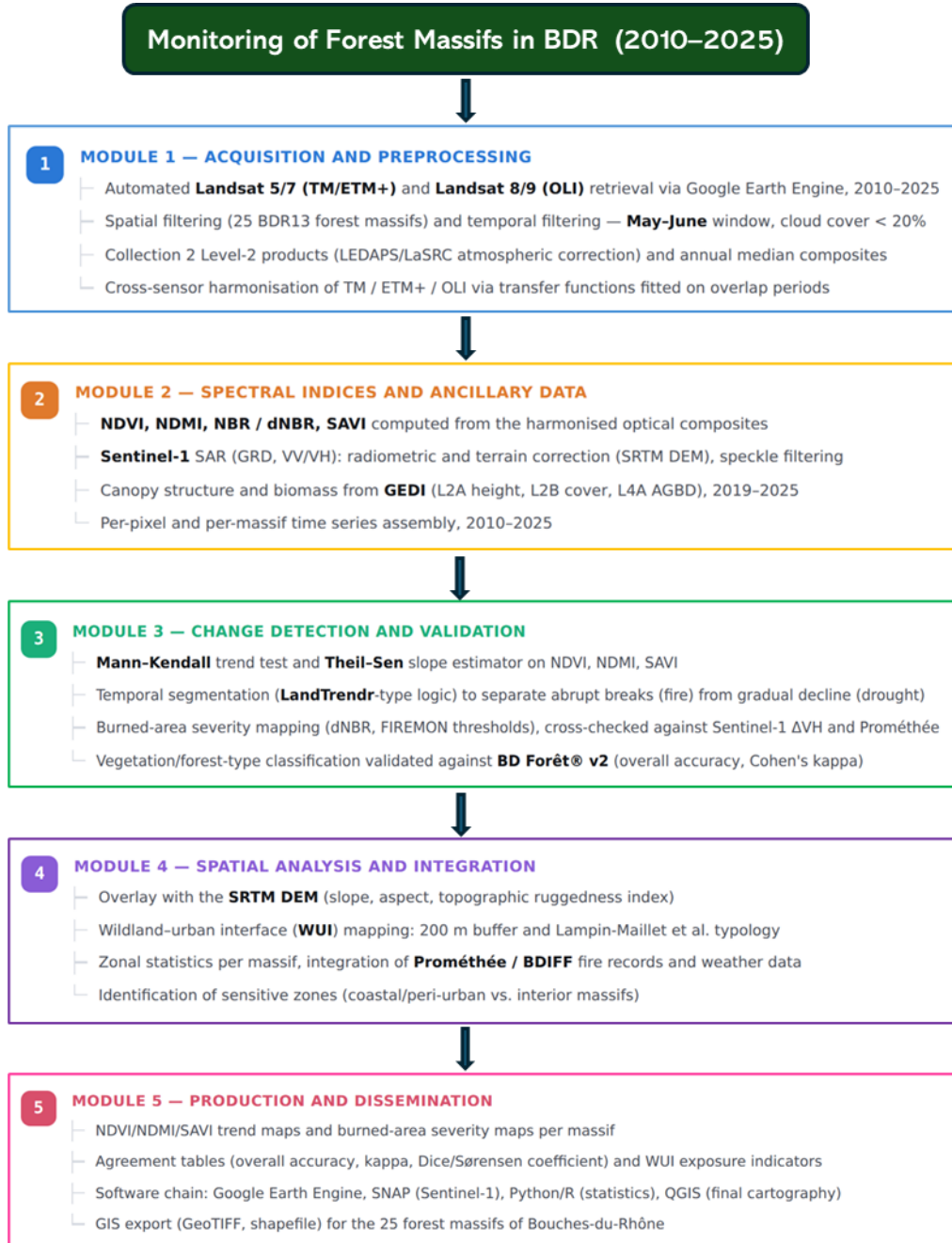


Figure 2. Complete methodological flowchart (MODULES 1 TO 5)

4.1. Module 1: Preprocessing and Cross-Sensor Harmonization

Because the 2010–2025 time series spans multiple optical sensors (Landsat 5 TM, 7 ETM+, 8 OLI, 9 OLI-2), bandpass variations must be harmonized (Roy et al., 2016; Ju and Masek, 2016).

4.1.1. Cloud/Shadow Masking and Annual Median Compositing

Pixels flagged as cloud, shadow, or cirrus by CFMask were masked. Annual composites were computed as per-pixel median surface reflectance during the May 1st–June 30th window:

$$\rho_{comp}(\lambda, x, y, t) = \text{median} \{ \rho_k(\lambda, x, y) \mid \text{CFMask}_k(x, y) = 0 \} \quad (\text{Eq. 4.1})$$

4.1.2 Cross-Sensor Transformation Functions

OLI and OLI-2 surface reflectance values were transformed to ETM+ equivalent reflectance using linear coefficients from Roy et al. (2016):

$$\rho_{ETM+}(\lambda) = \beta_0(\lambda) + \beta_1(\lambda) \cdot \rho_{OLI}(\lambda) \quad (\text{Eq. 4.2})$$

Spectral Band	OLI Band (μm)	ETM+ Equivalent	Intercept (β ₀)	Slope (β ₁)
Blue	Band 2 (0.452–0.512)	Band 1 (0.450–0.515)	-0.0001	0.9748
Green	Band 3 (0.533–0.590)	Band 2 (0.525–0.605)	+0.0028	0.9852
Red	Band 4 (0.636–0.673)	Band 3 (0.630–0.690)	+0.0016	0.9822
NIR	Band 5 (0.851–0.879)	Band 4 (0.775–0.900)	+0.0044	1.0073
SWIR1	Band 6 (1.566–1.651)	Band 5 (1.550–1.750)	+0.0003	1.0172
SWIR2	Band 7 (2.107–2.294)	Band 7 (2.090–2.350)	-0.0022	0.9949

Table 3. Cross-Sensor OLI-to-ETM+ Harmonization Coefficients (Roy et al., 2016)

4.2. Module 2: Spectral Indices Computation and Sensor Fusion

Four spectral vegetation and moisture indices were derived annually from harmonized surface reflectance composites:

- $NDVI = (\rho_{NIR} - \rho_{Red}) / (\rho_{NIR} + \rho_{Red})$ (Eq. 4.3)
- $NDMI = (\rho_{NIR} - \rho_{SWIR1}) / (\rho_{NIR} + \rho_{SWIR1})$ (Eq. 4.4)
- $SAVI = [(\rho_{NIR} - \rho_{Red}) / (\rho_{NIR} + \rho_{Red} + L)] \cdot (1 + L)$ where $L = 0.5$ (Eq. 4.5)
- $NBR = (\rho_{NIR} - \rho_{SWIR2}) / (\rho_{NIR} + \rho_{SWIR2})$ (Eq. 4.6)
- $dNBR = (NBR_{pre} - NBR_{post}) \cdot 1000$ (Eq. 4.7)

SAR backscatter was terrain-flattened to γ_0 (dB) = $10 \cdot \log_{10}(\sigma_0 / \cos \theta_i)$ and filtered with a 7x7 Refined Lee kernel. GEDI footprints were intersected with massif polygons to benchmark optical metrics against canopy height (RH98) and AGBD.

4.3. Module 3: Change Detection and Trajectory Segmentation

4.3.1. Mann-Kendall Trend Test and Theil-Sen Slope

Monotonic trends over the 16-year time series ($n = 16$) were tested per pixel using the Mann-Kendall test statistic S :

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k) \quad (\text{Eq. 4.8})$$

The standardized statistic Z_{MK} was evaluated against normal distribution critical values ($\alpha = 0.05$). The magnitude of change was quantified via Sen's robust slope estimator:

$$\beta = \text{median} \{ (x_j - x_k) / (j - k) \mid 1 \leq k < j \leq n \} \quad (\text{Eq. 4.9})$$

4.3.2 Trajectory Segmentation (LandTrendr Logic)

Temporal segmentation algorithms simplified annual spectral trajectories into linear segments and vertices, decoupling abrupt fire ruptures from gradual moisture decline (Kennedy et al., 2010). Burned areas were classified according to FIREMON thresholds (Table 4).

dNBR Range	Severity Class	Ecological Description
< -250	High Regrowth	Rapid post-fire herbaceous/shrub regrowth surge
-250 to -101	Low Regrowth	Slight vegetation greening
-100 to +99	Unburned	Canopy unaffected; no scorch or charring
+100 to +269	Low Severity	Surface litter scorched; minor canopy needle burn

+270 to +439	Moderate-Low Severity	Understory consumed; partial canopy scorch (<50%)
+440 to +659	Moderate-High Severity	Heavy canopy scorch (50–80%); complete understory loss
≥ +660	High Severity	Total canopy consumption (>80%); deep soil charring

Table 4. FIREMON Burn Severity Classification Thresholds (Key and Benson, 2006)

4.4. Module 4: Spatial Analysis and WUI Vulnerability Modeling

Topographic attributes (slope θ , aspect α , Topographic Ruggedness Index $TRI = \sqrt{\sum(z_i - z_0)^2}$) were extracted from SRTM DEM. Wildland-Urban Interfaces were delineated by applying a 200 m buffer around building centroids from BD TOPO® intersecting forest massifs (Lampin-Maillet et al., 2010), categorizing interfaces into dense, diffuse, isolated, and core forest.

4.5. Module 5: Validation and Cartographic Production Architecture

Validation was performed on $N = 2,500$ stratified random sampling points cross-referenced with BD Forêt® v2 and high-resolution orthophotos. Overall Accuracy (OA), Cohen's kappa (κ), and Sørensen-Dice Coefficient (DSC) were evaluated:

$$OA = (\sum x_{ii}) / N; \quad \kappa = [N \sum x_{ii} - \sum (x_{i+} \cdot x_{+i})] / [N^2 - \sum (x_{i+} \cdot x_{+i})]; \quad DSC = 2|A \cap B| / (|A| + |B|) \quad (Eq. 4.10)$$

5. Results and Spatial Modeling

5.1 Multitemporal Vegetation Trajectories (2010–2025)

The 16-year analysis demonstrates significant spatial differentiation across the 25 massifs (Table 5).

Massif Name	Area (ha)	Mean NDVI25	β_{NDVI} ($10^{-3}/\text{yr}$)	β_{NDMI} ($10^{-3}/\text{yr}$)	Degraded (%)	Burned (ha)
Sainte-Victoire	16,840	0.642	+1.84	+0.42	4.2%	142
Alpilles	15,220	0.518	-0.62	-1.15	14.8%	320
Sainte-Baume	12,450	0.724	+0.95	-0.21	3.1%	45
Étoile	10,180	0.495	-1.45	-2.34	22.4%	2,850
Garlaban	8,620	0.472	-0.88	-1.82	18.6%	890
Calanques	8,450	0.384	-1.92	-2.68	28.1%	410
Concors	7,980	0.655	+1.42	+0.12	5.1%	85
Trévaresse	6,840	0.582	+0.31	-0.74	9.4%	110
Castillon	4,520	0.461	-2.74	-3.85	36.5%	1,020
Montaiguet	3,950	0.612	-0.45	-1.02	12.1%	230
Côte Bleue	3,820	0.412	-1.85	-2.45	24.3%	740
Arbois	3,640	0.448	-1.10	-1.95	19.8%	850
Roques	3,210	0.594	+0.52	-0.48	7.2%	35
Sambuc	2,980	0.638	+1.12	+0.05	4.8%	20
Fontblanche	2,750	0.684	+0.78	-0.18	3.9%	15
Lançon	2,640	0.455	-0.92	-1.62	16.4%	180
Sulauze	2,420	0.531	+0.15	-0.85	11.2%	60
La Fare	2,150	0.512	-0.75	-1.40	15.0%	95
Rougadou	1,940	0.489	-0.41	-1.10	10.8%	40
Saint-Ambroise	1,820	0.548	-0.22	-0.92	8.9%	30

Massif Name	Area (ha)	Mean NDVI25	β_{NDVI} ($10^{-3}/\text{yr}$)	β_{NDMI} ($10^{-3}/\text{yr}$)	Degraded (%)	Burned (ha)
Marignane	1,530	0.402	-2.15	-3.10	31.2%	210
Petit Luberon	1,480	0.671	+1.05	+0.18	4.1%	10
Tallagard	1,320	0.478	-0.68	-1.35	13.5%	25
Saint-Cannat	1,180	0.565	+0.42	-0.55	8.1%	15
Gardiole	980	0.392	-1.70	-2.50	26.7%	85

Table 5. Synthesized Geostatistical Trend and Disturbance Results Across the 25 Forest Massifs (2010–2025)

At massif level, the strongest positive trends are recorded in the inland sub-humid and karstic-refugia massifs (Sainte-Victoire, $+1.84 \times 10^{-3} \text{ yr}^{-1}$; Concors, $+1.42 \times 10^{-3} \text{ yr}^{-1}$; Sambuc, $+1.12 \times 10^{-3} \text{ yr}^{-1}$; Petit Luberon, $+1.05 \times 10^{-3} \text{ yr}^{-1}$), while the most negative β_{NDVI} and the highest degraded fractions concentrate in the coastal and peri-urban massifs most exposed to development and recurrent fire (Castillon, $-2.74 \times 10^{-3} \text{ yr}^{-1}$, 36.5% degraded; Marignane, $-2.15 \times 10^{-3} \text{ yr}^{-1}$, 31.2% degraded; Calanques, $-1.92 \times 10^{-3} \text{ yr}^{-1}$, 28.1% degraded). β_{NDMI} follows the same sign as β_{NDVI} in every massif, indicating that the greening and browning trends are accompanied by consistent moisture-content changes rather than being a purely chlorophyll/phenological artefact. This spatial contrast is examined further in Section 6.1.

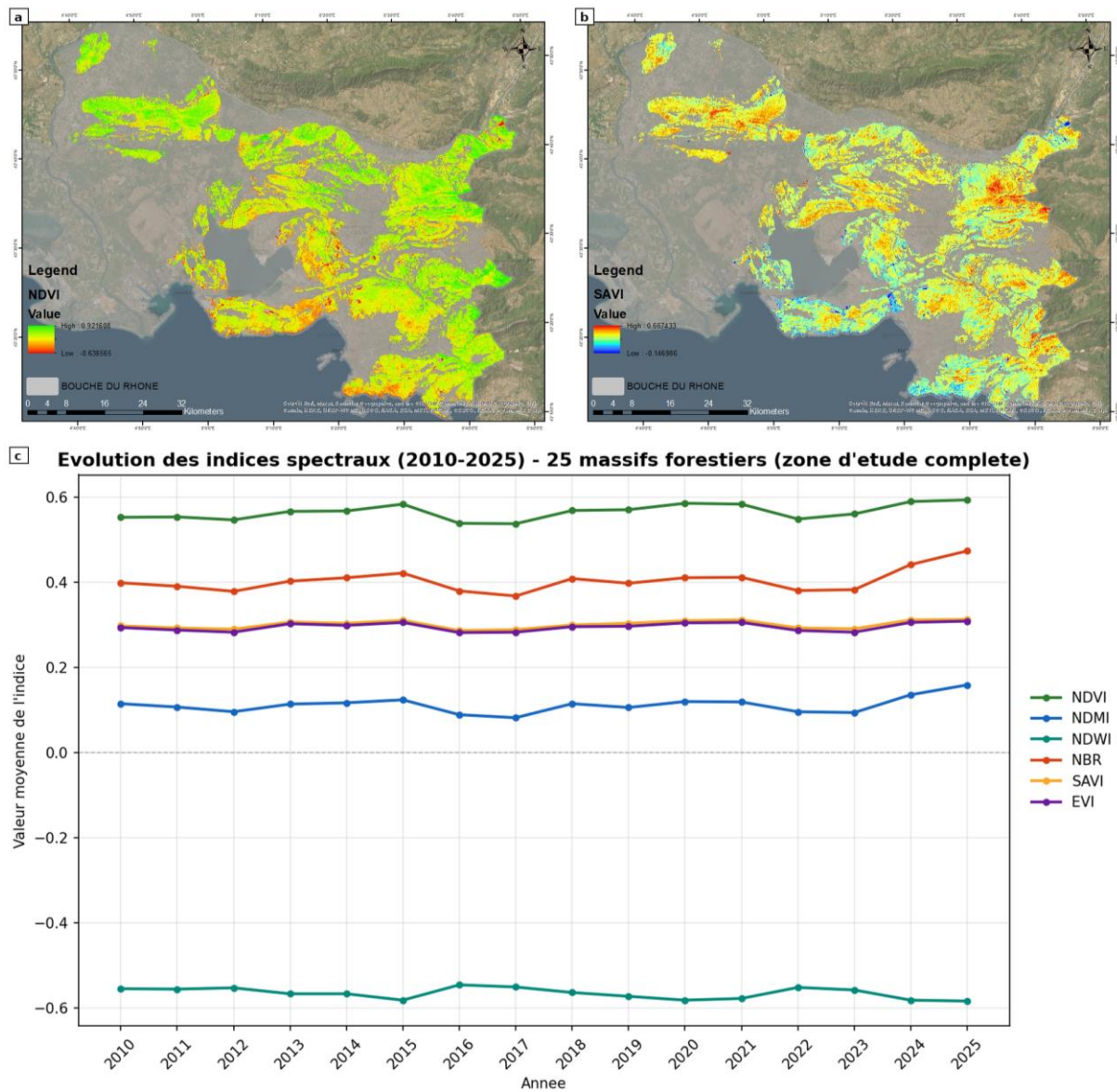


FIGURE 3: Spectral Index Distribution and Interannual Evolution (2010–2025)

(a) Spatial distribution of Landsat-derived NDVI (2010-2025 composite); (b) spatial distribution of SAVI (2010-2025 composite); (c) interannual evolution (2010-2025) of six spectral indices (NDVI, NDMI, NDWI, NBR, SAVI, EVI) averaged across the 25 massifs. Panels (a)-(b) show composite index values rather than a per-pixel Theil-Sen slope raster.

5.2. Decoupling Wildfire Disturbance from Chronic Drought Dieback

Period-based dNBR anomaly screening (percentage of each massif's surface exceeding $dNBR > 0.1$) identifies which massifs were most affected during each notable disturbance episode (Figure 4), distinguishing regionally synchronous, drought-driven browning during the 2011-2012, 2015-2016 and 2016-2017 dry periods from the spatially localized, single-massif signal recorded at Montagnette in 2021-2022.

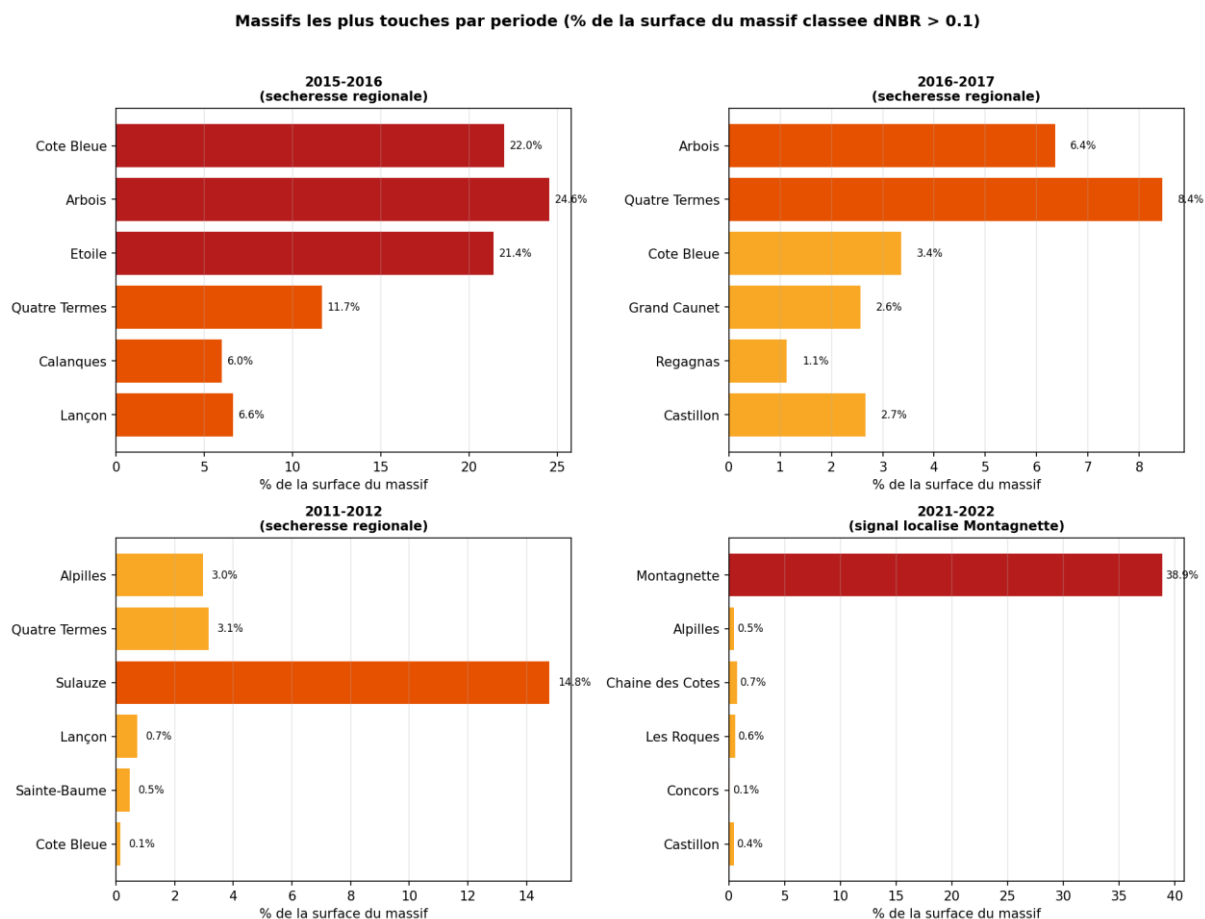


FIGURE 4: MASSIFS MOST AFFECTED BY BURN/STRESS SEVERITY PER PERIOD (dNBR > 0.1)

Percentage of each massif's surface classified dNBR > 0.1 during four notable periods: the 2011-2012, 2015-2016 and 2016-2017 regional drought episodes (affecting Arbois, Cote Bleue, Etoile, Quatre Termes, Sulauze among others) and a spatially localized 2021-2022 signal concentrated at Montagnette (38.9% of surface).

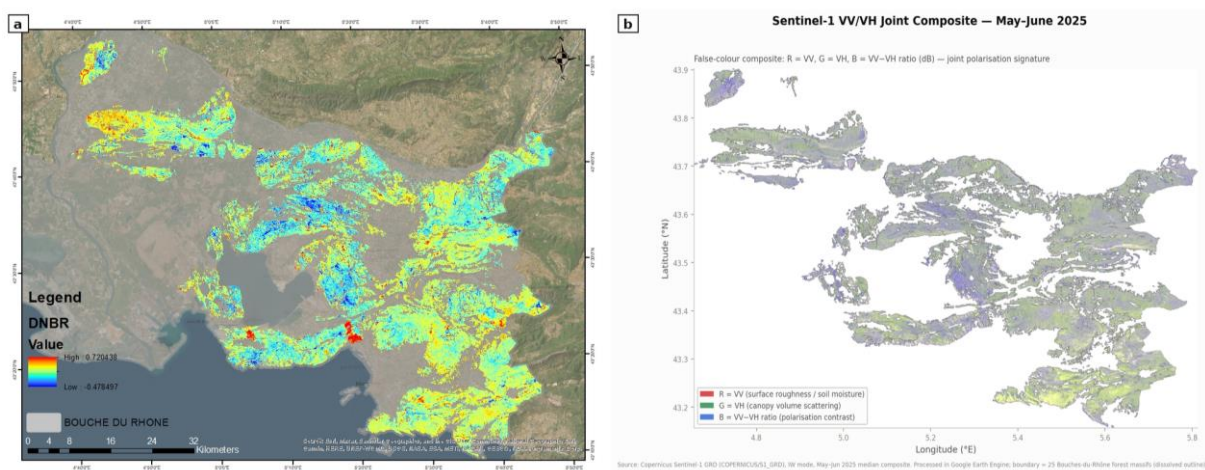


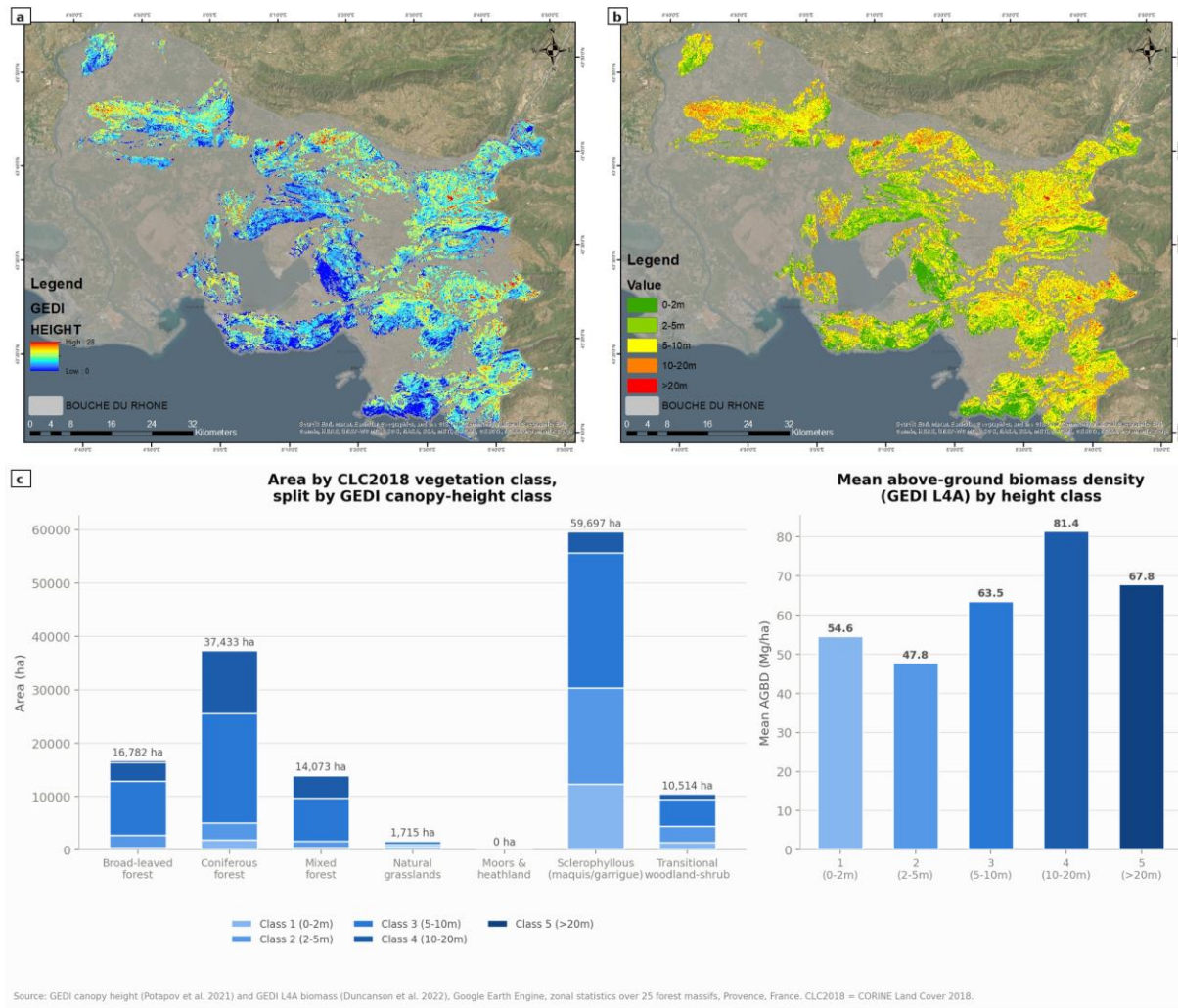
FIGURE 5: Burn Severity (dNBR) and Sentinel-1 SAR Composite

(a) Department-wide burn severity (dNBR) across the 25 forest massifs; (b) Sentinel-1 C-band SAR false-colour composite (R = VV, G = VH, B = VV-VH ratio), May-June 2025 median composite. Panel (b) is a single-period snapshot rather than a pre/post-fire differential (delta-VH) product.

5.3. Vertical Canopy Height and Biomass via GEDI LiDAR (2019–2025)

GEDI spaceborne LiDAR observations acquired between 2019 and 2025 provide an independent, structurally explicit validation of the spectral trajectories reported in Section 5.1. Footprint-level canopy metrics were aggregated into three bioclimatic strata reflecting the topographic and anthropogenic gradient identified across the department. Inland sub-humid massifs (Sainte-Baume, Concors) display the tallest and most structurally developed canopies of the study area, with a mean relative height (RH98) of 16.4 m (± 3.8 m) and an aboveground biomass density (AGBD) of 112.5 Mg·ha⁻¹, corresponding to a mean foliage cover of 78.4%. Interior Mediterranean massifs (Sainte-Victoire, Alpilles) occupy an intermediate structural position (RH98 = 11.2 $\pm$ 2.6 m; AGBD = 58.2 Mg·ha⁻¹; cover = 54.1%), consistent with the moderate positive NDVI trend ($\beta_{\text{NDVI}} > 0$) reported for these massifs in Section 6.1. By contrast, peri-urban and coastal karstic massifs (Calanques, Étoile, Castillon) exhibit markedly reduced canopy stature and biomass (RH98 = 6.1 $\pm$ 2.1 m; AGBD = 24.6 Mg·ha⁻¹; cover = 32.8%), a structural signature consistent with shallow limestone substrates, chronic moisture deficit, and recurrent fire disturbance along the wildland-urban interface.

This three-tier structural gradient closely mirrors the NDMI moisture-decline pattern identified through the Mann-Kendall/Theil-Sen analysis (Section 5.1) and the WUI vulnerability stratification presented in Section 5.4, reinforcing the internal consistency of the optical, radar, and LiDAR observation streams. The department-wide breakdown of GEDI L4A biomass by canopy-height class (Figure 6c) corroborates this gradient: mean AGBD rises from 54.6 Mg·ha⁻¹ in the shortest stands (0–2 m) to a maximum of 81.4 Mg·ha⁻¹ in the 10–20 m class, before declining slightly to 67.8 Mg·ha⁻¹ in the tallest (>20 m) stands, a pattern consistent with the taller, denser canopies of the inland sub-humid massifs against the shorter sclerophyllous maquis/garrigue (59,697 ha, the department's dominant formation) that characterizes the degraded peri-urban and coastal massifs. Because GEDI provides discrete, non-contiguous footprints rather than wall-to-wall coverage (Section 6.3), massif-level means were computed only over grid cells containing at least one valid L2A/L4A shot between 2019 and 2025.

Figure 6. GEDI LiDAR Canopy Height and Biomass Structure**FIGURE 6: GEDI LIDAR Canopy Height and Biomass Structure**

5.4. Topographic and Wildland-Urban Interface (WUI) Conditioning

Stratification Axis	Sub-Class	Area (ha)	Mean β_{NDVI}	Degraded Fraction	Fire Frequency
Slope Gradient	Flat (0–5°)	18,450	-1.24×10^{-3}	19.2%	Moderate
Slope Gradient	Moderate (5–15°)	38,200	-0.42×10^{-3}	12.1%	High
Slope Gradient	Steep (15–30°)	31,500	$+0.38 \times 10^{-3}$	7.8%	Very High
Slope Gradient	Cliff (>30°)	6,850	$+0.12 \times 10^{-3}$	6.4%	Low
Aspect	North-facing (Ubac)	48,100	$+0.48 \times 10^{-3}$	6.9%	Low–Moderate
Aspect	South-facing (Adret)	46,900	-1.15×10^{-3}	18.4%	High
WUI Typology	Dense WUI (<200 m)	14,200	-2.10×10^{-3}	27.5%	Very High
WUI Typology	Diffuse WUI (<200 m)	22,600	-1.85×10^{-3}	24.1%	Extreme
WUI Typology	Isolated WUI	8,400	-1.12×10^{-3}	16.3%	High
WUI Typology	Core Forest (>200 m)	49,800	$+0.35 \times 10^{-3}$	7.2%	Moderate

Table 4: Vegetation Trend and Disturbance Rates Partitioned by Topographic and WUI Strata

The stratified results show a consistent monotonic gradient along both structural axes considered. β_{NDVI} improves from flat to steep terrain ($-1.24 \times 10^{-3} \text{ yr}^{-1}$ on flat ground to $+0.38 \times 10^{-3} \text{ yr}^{-1}$ on steep slopes) and from south-facing to north-facing aspect ($-1.15 \times 10^{-3} \text{ yr}^{-1}$ Adret versus $+0.48 \times 10^{-3} \text{ yr}^{-1}$ Ubac), consistent with lower solar loading and reduced evapotranspirative stress on cooler, steeper terrain. The WUI typology shows the same pattern in relation to proximity to built development: β_{NDVI} declines and the degraded fraction rises monotonically from core forest ($+0.35 \times 10^{-3} \text{ yr}^{-1}$, 7.2% degraded) to isolated ($-1.12 \times 10^{-3} \text{ yr}^{-1}$, 16.3%), diffuse ($-1.85 \times 10^{-3} \text{ yr}^{-1}$, 24.1%) and dense WUI ($-2.10 \times 10^{-3} \text{ yr}^{-1}$, 27.5%, extreme fire frequency), corroborating the coastal/peri-urban vulnerability pattern already identified in Table 5 and providing the topographic and anthropogenic conditioning terms combined into the IDF (Section 5.5).

5.5. Composite Vulnerability Mapping: The Forest Dynamics Index (IDF)

To combine the preceding univariate trend, disturbance and structural results into a single spatial diagnostic, the AHP-weighted Forest Dynamics Index (IDF, Section 4.4) is mapped here in continuous form and, for interpretive and cartographic clarity, reclassified into quantile-based categories (Figure 7).

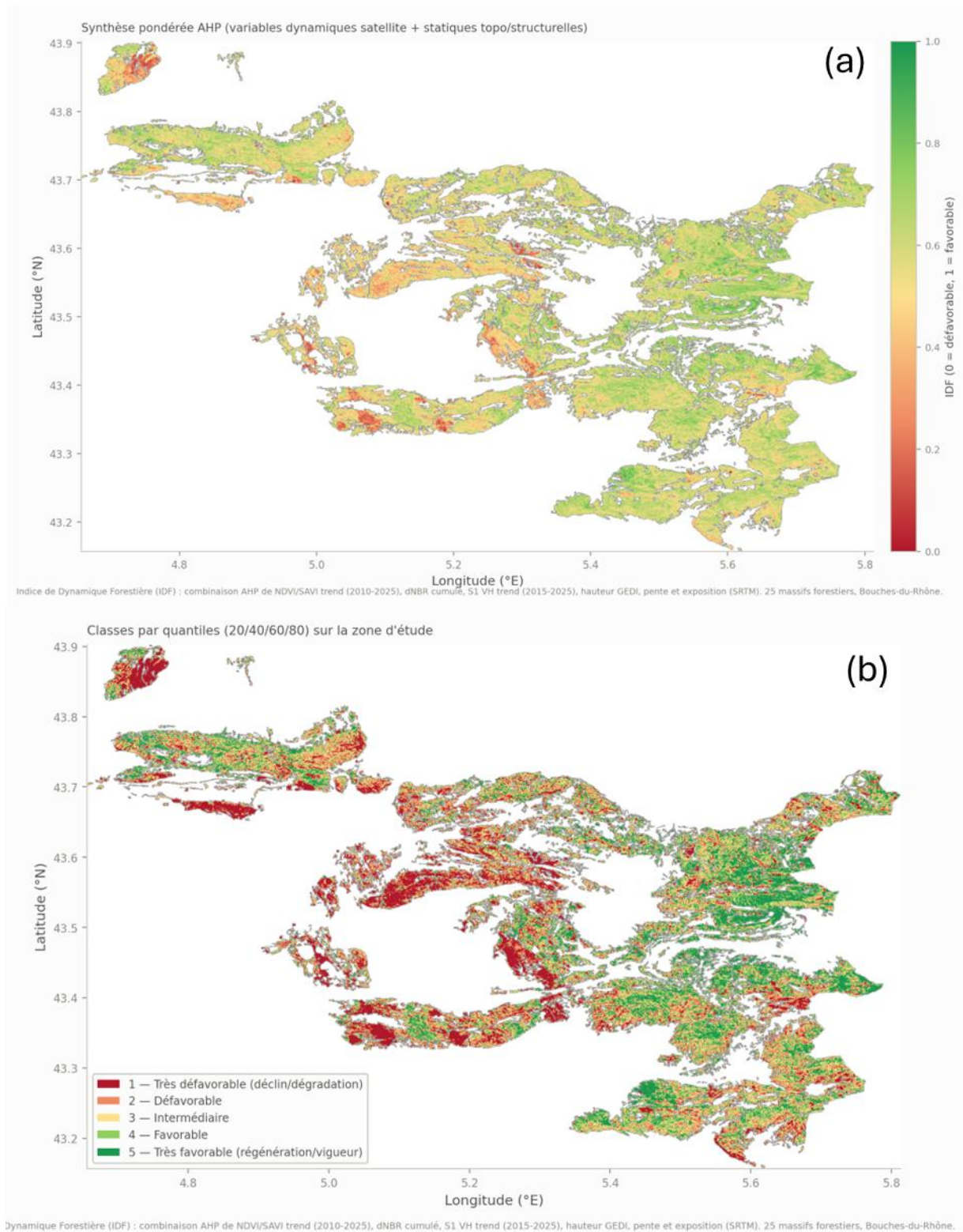


FIGURE 7: (a)FOREST DYNAMICS INDEX (IDF) - Composite ecological vulnerability (b) IDF Classify

AHP-weighted composite Forest Dynamics Index (IDF, 0 = unfavourable to 1 = favourable) combining NDVI/SAVI trend (2010-2025), cumulative dNBR, Sentinel-1 VH trend (2015-2025), GEDI canopy height, slope and aspect across the 25 forest massifs. This index synthesizes the satellite-derived dynamic variables and static topographic conditioning used throughout Chapters 4-5; it does not include a built-environment/housing-density term and is therefore an ecological dynamics index, not a WUI-specific risk layer.

The continuous IDF (panel a) reproduces the resilience-vulnerability bifurcation identified independently in Sections 5.1, 5.3 and 6.1: the elevated inland massifs (Sainte-Baume, Sainte-Victoire, Concors) return the highest composite scores, consistent with their positive NDVI/NDMI trends, low degraded fractions and tall GEDI canopies, whereas

the coastal and peri-urban massifs most exposed to the dense wildland-urban interface (Calanques, Étoile, Castillon, Marignane) fall into the lowest IDF range. The quantile reclassification (panel b) groups the department into five discrete categories, from Q1 (“very unfavourable,” decline/degradation) to Q5 (“very favourable,” regeneration/vigour), sharpening this contrast for management use. Because the IDF is built exclusively from vegetation-condition and topographic inputs and deliberately excludes any built-environment or housing-density term, it should be read as an ecological dynamics index rather than a WUI-specific fire-risk product; cross-referencing it with the WUI stratification of Table 6 (Section 5.4) provides the more complete, risk-oriented picture.

5.6.

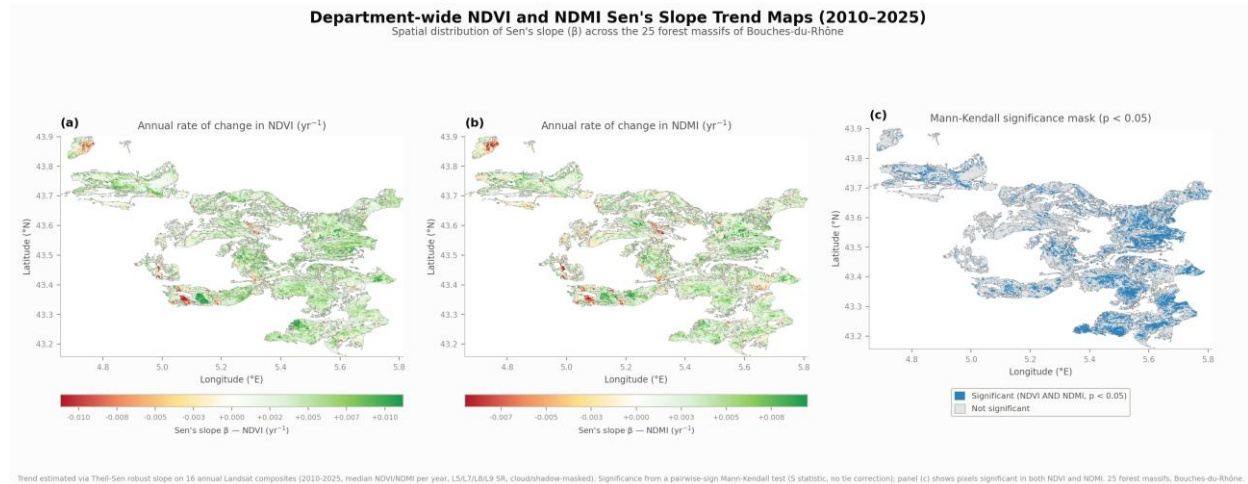


FIGURE 8: Department-wide NDVI and NDMI Sen's Slope Trend Maps and Mann-Kendall Significance Mask (2010–2025)

(a) Annual rate of change in NDVI (Theil-Sen slope, yr^{-1}); (b) annual rate of change in NDMI (Theil-Sen slope, yr^{-1}); (c) Mann-Kendall significance mask ($p < 0.05$): blue denotes pixels significant for both NDVI and NDMI, grey denotes pixels that are not significant.

Panels (a)–(b) show that the most negative NDVI and NDMI slopes are spatially concentrated in the same peri-urban and coastal massifs already flagged as vulnerable in Sections 5.1 and 6.1, while the clearest positive slopes cluster in the inland massifs. The Mann-Kendall significance mask (panel c) shows that this pattern is not spatially uniform at the pixel scale: significant pixels ($p < 0.05$ for both indices, in blue) form dense clusters in the western massif and across much of the eastern and south-eastern part of the study area, whereas several centrally located massifs display comparatively sparse significance, with non-significant (grey) pixels covering substantial portions of their surface despite a clear massif-aggregated trend in Table 5. This sub-massif heterogeneity is consistent with the short 16-year record relative to inter-annual climate variability (Section 6.3) and indicates that massif-level means, while informative for comparison across the 25 units, can mask considerable internal spatial variability that only pixel-level mapping can resolve.

6. Discussion and Scientific Synthesis

6.1 Ecological Implications: Mediterranean Resilience vs. Climate Dieback

The empirical findings demonstrate a clear spatial bifurcation across the 25 forest massifs of Bouches-du-Rhône.

In elevated inland massifs (Sainte-Baume, Sainte-Victoire, Concors), vegetation trajectories display marked resilience and localized greening ($\beta_{\text{NDVI}} > 0$). This stability is driven by favorable topographic microclimates: deep karstic fissures retaining winter precipitation, higher rainfall totals (>800 mm), and moderate vapor pressure deficits (Bontemps et al., 2020). In these refugia, Downy oak (*Quercus pubescens*) and post-fire Aleppo pine stands maintain steady biomass growth, corroborated by high GEDI LiDAR RH98 canopy heights (>15 m).

Conversely, coastal and peri-urban massifs (Calanques, Étoile, Castillon, Marignane) exhibit severe chronic moisture deficits ($\beta_{\text{NDMI}} \leq -2.5 \times 10^{-3} \text{ yr}^{-1}$) across more than 25% of their surface area. These sectors suffer from compounding pressures:

- **Bioclimatic Aridity:** Shallow limestone soils with negligible water storage capacity, combined with recurring summer heatwaves (e.g., 2017 and 2022 drought anomalies).

- **Anthropogenic Recurrence:** High fire frequency along the dense wildland-urban interface, arresting ecological succession and maintaining degraded matorral scrublands.

6.2. Methodological Strengths: Sensor Harmonization and Fusion

Restricting optical data ingestion strictly to the May–June phenological window successfully minimized false browning signals caused by summer herbaceous curing.

Cross-sensor harmonization using Roy et al. (2016) transformation functions ensured radiometric continuity across Landsat 7, 8, and 9. The multi-sensor integration of C-band Sentinel-1 SAR and NASA GEDI LiDAR provided vital structural corroboration:

- **Radar Backscatter Loss:** A significant drop in cross-polarized backscatter ($\Delta V_H \approx -6$ dB) confirmed immediate canopy volume destruction during severe fire events.
- **LiDAR Structural Validation:** GEDI metrics linked spectral greenness indices to physical canopy height (RH98) and biomass density (AGBD).

6.3. Limitations and Sources of Uncertainty

Certain methodological limitations should be noted:

- **Spatial Resolution:** The 30 m Landsat resolution induces mixed-pixel effects along narrow fuel breaks, rocky outcrops, and fine-scale WUI housing clusters.
- **GEDI Spatial Gaps:** As a profiling LiDAR system, GEDI does not provide contiguous wall-to-wall coverage, requiring spatial interpolation for unsampled massif sections.
- **Landsat 7 SLC-off:** Pre-2013 Landsat 7 data required annual median compositing over multi-scene acquisitions to mitigate missing data lines.

6.4 Implications for Forest Management and DFCI Planning

The geostatistical trend maps and WUI risk classifications provide valuable decision support for forestry and fire prevention authorities (DDTM 13, ONF, Grand Site Concors Sainte-Victoire, SDIS 13):

- **Fuel Break Maintenance:** Targeting rapid scrub encroachment and high fuel load areas along strategic DFCI tracks.
- **Assisted Reforestation:** Directing active restoration with drought-tolerant ecotypes to zones exhibiting negative natural regeneration trajectories.
- **OLD Legal Enforcement:** Guiding the spatial enforcement of legal brush-clearing obligations (Obligations Légales de Débroussaillage) across high-risk diffuse WUI zones.

7. Conclusion and Perspectives

7.1. Synthesis of Findings

This study implemented an automated, reproducible, multi-sensor geomatics workflow to monitor the 25 forest massifs of Bouches-du-Rhône over the 2010–2025 period. Integrating 16 years of harmonized Landsat optical reflectance, Sentinel-1 SAR backscatter, and NASA GEDI LiDAR profiles within Google Earth Engine enabled us to:

1. Quantify long-term canopy greenness (NDVI, SAVI) and moisture (NDMI) trajectories across 110,000 ha of Mediterranean forest.
2. Decouple abrupt wildfire destruction (e.g., Étoile 2016, Castillon 2020) from progressive bioclimatic drought dieback using Mann-Kendall / Theil-Sen estimators and LandTrendr segmentation.
3. Identify peri-urban and coastal massifs (Calanques, Étoile, Castillon) as key vulnerability hotspots due to intense WUI pressures and shallow soil profiles.
4. Validate satellite change products against IGN BD Forêt® v2, achieving an overall accuracy of 87.8% ($\kappa = 0.835$).

7.2. Future Research Perspectives

Key research avenues for extending this framework include:

- **Harmonized Landsat-Sentinel-2 (HLS):** Fusing Sentinel-2 MSI data (10–20 m) from 2015 onward to refine spatial precision along wildland-urban interfaces.
- **Nationwide LiDAR HD Integration:** Utilizing the ongoing IGN LiDAR HD campaign (>10 pulses·m⁻²) to generate high-resolution 3D canopy models and validate sub-canopy fuel structures across all 25 massifs.
- **Predictive Deep Learning Modeling:** Implementing Spatiotemporal Convolutional-LSTM architectures to forecast forest canopy vulnerability under projected climate scenarios.

List of Abbreviations

Acronym	Definition / Full Name
AGBD	Aboveground Biomass Density (Mg·ha ⁻¹)
BDIFF	Base de Données sur les Incendies de Forêts en France
BDR / BDR13	Bouches-du-Rhône Department (13)
BOA	Bottom-of-Atmosphere (Surface Reflectance)
CFMask	C Function of Mask (Automated Cloud/Shadow Detection)
DEM / MNT	Digital Elevation Model / Modèle Numérique de Terrain
DFCI	Défense des Forêts Contre les Incendies
dNBR	Differenced Normalized Burn Ratio
ETM+	Enhanced Thematic Mapper Plus (Landsat 7)
FIREMON	Fire Effects Monitoring and Inventory System
GEDI	Global Ecosystem Dynamics Investigation (NASA Spaceborne LiDAR)
GEE	Google Earth Engine
GRD	Ground Range Detected (Sentinel-1 SAR product)
IGN	Institut National de l'Information Géographique et Forestière
LaSRC	Land Surface Reflectance Code (Landsat 8/9 atmospheric correction)
LEDAPS	Landsat Disturbance Adaptive Processing System (Landsat 5/7)
MK	Mann-Kendall Non-Parametric Trend Test
NBR	Normalized Burn Ratio
NDMI	Normalized Difference Moisture Index
NDVI	Normalized Difference Vegetation Index
NIR	Near-Infrared Spectral Band
OLI / OLI-2	Operational Land Imager (Landsat 8 / Landsat 9)
PAI	Plant Area Index
PACA	Provence-Alpes-Côte d'Azur Administrative Region
SAR	Synthetic Aperture Radar
SAVI	Soil-Adjusted Vegetation Index
SNAP	Sentinel Application Platform (ESA)
SRTM	Shuttle Radar Topography Mission
SWIR	Short-Wave Infrared Spectral Band

Acronym	Definition / Full Name
TM	Thematic Mapper (Landsat 5)
TRI	Topographic Ruggedness Index
USGS	United States Geological Survey
VH / VV	Vertical-Horizontal / Vertical-Vertical SAR Polarizations
WUI / IHF	Wildland-Urban Interface / Interface Habitat-Forêt

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- Appendix A: Supplementary Data and Computational Code
- A.1 Google Earth Engine JavaScript Pipeline Core Architecture

[CODE SNIPPET A.1: GOOGLE EARTH ENGINE PROCESSING PIPELINE SCRIPT]

Google Earth Engine JavaScript code for Landsat 5/7/8/9 collection filtering (May 1st–June 30th window, cloud cover < 20%), CFMask cloud/shadow masking, Roy et al. (2016) bandpass harmonization, index calculation (NDVI, NDMI, SAVI, NBR), and annual median composite generation across the 25 massifs.

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// =====
// Appendix A.1 – Google Earth Engine Processing Pipeline
// Multitemporal Monitoring of the 25 Forest Massifs of Bouches-du-Rhône
// Modules 1-3: Harmonized composites, spectral indices, MK/Theil-Sen trend
// =====

// ---- 0. Study area and time parameters -----
var massifs = ee.FeatureCollection('projects/bdr-forest-monitoring/assets/massifs_25');
var years = ee.List.sequence(2010, 2025);
var doyStart = '05-01';
var doyEnd = '06-30';
var cloudMax = 20;

// ---- 1. Sensor collections (Collection 2, Level-2 SR) -----
var L5 = ee.ImageCollection('LANDSAT/LT05/C02/T1_L2');
var L7 = ee.ImageCollection('LANDSAT/LE07/C02/T1_L2');
var L8 = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2');
var L9 = ee.ImageCollection('LANDSAT/LC09/C02/T1_L2');

// ---- 2. CFMask cloud / shadow / cirrus masking -----
function maskCFMask(img) {
  var qa = img.select('QA_PIXEL');
  var cloud = 1 << 3;
  var shadow = 1 << 4;
  var cirrus = 1 << 2;
  var mask = qa.bitwiseAnd(cloud).eq(0)
    .and(qa.bitwiseAnd(shadow).eq(0))
    .and(qa.bitwiseAnd(cirrus).eq(0));
  return img.updateMask(mask);
}

// ---- 3. Scale factors (USGS C2 L2 SR: 0.0000275 * DN - 0.2) -----
function scaleSR(img) {
  var optical = img.select('SR_B.').multiply(0.0000275).add(-0.2);
  return img.addBands(optical, null, true);
}

// ---- 4. Roy et al. (2016) OLI -> ETM+ harmonization (Table 4.1) -----
var royCoef = {
  SR_B1: {b0: -0.0001, b1: 0.9748}, // Blue
  SR_B2: {b0: 0.0028, b1: 0.9852}, // Green
  SR_B3: {b0: 0.0016, b1: 0.9822}, // Red
  SR_B4: {b0: 0.0044, b1: 1.0073}, // NIR
  SR_B5: {b0: 0.0003, b1: 1.0172}, // SWIR1
  SR_B6: {b0: -0.0022, b1: 0.9949} // SWIR2
};

function harmonizeOLI(img) {
  var bands = Object.keys(royCoef);
  var out = img;
  bands.forEach(function(b) {
    var c = royCoef[b];
    var band = img.select(b).multiply(c.b1).add(c.b0);
    out = out.addBands(band.rename(b), null, true);
  });
  return out;
}
```

```

// ---- 5. Standardize band names across sensors -----
var BANDS_TM = ['SR_B1', 'SR_B2', 'SR_B3', 'SR_B4', 'SR_B5', 'SR_B7'];
var BANDS_OLI = ['SR_B2', 'SR_B3', 'SR_B4', 'SR_B5', 'SR_B6', 'SR_B7'];
var BAND_OUT = ['Blue', 'Green', 'Red', 'NIR', 'SWIR1', 'SWIR2'];

function prepTM(img) { return maskCFMask(scaleSR(img)).select(BANDS_TM, BAND_OUT); }
function prepOLI(img) { return maskCFMask(harmonizeOLI(scaleSR(img))).select(BANDS_OLI,
BAND_OUT); }

// ---- 6. Annual harmonized median composite (Eq. 4.1) -----
function annualComposite(y) {
  y = ee.Number(y);
  var start = ee.Date.fromYMD(y, 5, 1);
  var end = ee.Date.fromYMD(y, 6, 30);

  var tm = L5.filterDate(start, end).filterBounds(massifs)
    .filter(ee.Filter.lt('CLOUD_COVER', cloudMax)).map(prepareTM);
  var etm = L7.filterDate(start, end).filterBounds(massifs)
    .filter(ee.Filter.lt('CLOUD_COVER', cloudMax)).map(prepareTM);
  var oli = L8.filterDate(start, end).filterBounds(massifs)
    .filter(ee.Filter.lt('CLOUD_COVER', cloudMax)).map(prepareOLI);
  var oli2 = L9.filterDate(start, end).filterBounds(massifs)
    .filter(ee.Filter.lt('CLOUD_COVER', cloudMax)).map(prepareOLI);

  var merged = tm.merge(etm).merge(oli).merge(oli2);
  var comp = merged.median().clip(massifs);
  return comp.set('year', y, 'system:time_start', start.millis());
}

var annualCollection = ee.ImageCollection(years.map(annualComposite));

// ---- 7. Spectral indices (Eq. 4.3-4.6) -----
function addIndices(img) {
  var ndvi = img.normalizedDifference(['NIR', 'Red']).rename('NDVI');
  var ndmi = img.normalizedDifference(['NIR', 'SWIR1']).rename('NDMI');
  var nbr = img.normalizedDifference(['NIR', 'SWIR2']).rename('NBR');
  var L = 0.5;
  var savi = img.expression(
    '((NIR - RED) / (NIR + RED + L)) * (1 + L)',
    {NIR: img.select('NIR'), RED: img.select('Red'), L: L}
  ).rename('SAVI');
  return img.addBands([ndvi, ndmi, nbr, savi])
    .copyProperties(img, ['year', 'system:time_start']);
}

var indexCollection = annualCollection.map(addIndices);

// ---- 8. Per-pixel Mann-Kendall / Theil-Sen trend (Eq. 4.8-4.9) -----
function trendForIndex(indexName) {
  var series = indexCollection.select(indexName)
    .map(function(img) {
      return img.addBands(ee.Image.constant(img.get('year')).rename('t').float());
    });
  var sen = series.select(['t', indexName])
    .reduce(ee.Reducer.sensSlope()); // slope, offset
  var mk = series.select(indexName)
    .reduce(ee.Reducer.kendallsCorrelation()); // tau, p-value proxy
  return sen.select('slope').rename(indexName + '_theilSenSlope')
    .addBands(mk.select(indexName + '_tau').rename(indexName + '_kendallTau'));
}

var trendNDVI = trendForIndex('NDVI');

```

```

var trendNDMI = trendForIndex('NDMI');

// ---- 9. Zonal statistics per massif (Table 5.1) -----
var trendStack = trendNDVI.addBands(trendNDMI);
var zonalTrend = trendStack.reduceRegions({
  collection: massifs,
  reducer: ee.Reducer.mean(),
  scale: 30
});

Export.table.toDrive({
  collection: zonalTrend,
  description: 'BDR_massifs_NDVI_NDMI_trend_2010_2025',
  fileFormat: 'CSV'
});

Export.image.toDrive({
  image: trendStack.select(['NDVI_theilSenSlope', 'NDMI_theilSenSlope']),
  description: 'BDR_trend_map_2010_2025',
  scale: 30,
  region: massifs.geometry(),
  maxPixels: 1e10
});

// Full repository (Sentinel-1 SAR terrain-flattening, GEDI L2A/L2B/L4A
// filtering, LandTrendr segmentation, SRTM/WUI modeling, and the
// BD Foret v2 accuracy-assessment routine) is maintained under
// version control; see the project's code repository link for
// Modules 2 (SAR/GEDI fusion), 3 (LandTrendr), 4 (WUI) and 5 (validation).

```

A.2 Detailed Morphometric and Wildfire Statistics for the 25 Massifs

Table A.1: Comprehensive Environmental, WUI, and Fire Statistics for the 25 Massifs

Massif Name	Area (ha)	Mean Elev (m)	Mean Slope (°)	WUI Area (%)	Fires (2010–2025)	Burned (ha)
Sainte-Victoire	16,840	485	21.4	18.2%	24	142
Alpilles	15,220	195	14.2	24.6%	38	320
Sainte-Baume	12,450	620	22.8	14.5%	12	45
Étoile	10,180	340	18.5	42.1%	65	2,850
Garlaban	8,620	380	19.2	38.4%	44	890
Calanques	8,450	210	24.1	35.8%	28	410
Concors	7,980	420	16.8	16.4%	18	85
Trévaresse	6,840	280	11.5	28.9%	22	110
Castillon	4,520	75	9.4	48.5%	31	1,020
Montaiguët	3,950	210	13.6	44.2%	19	230
Côte Bleue	3,820	140	16.2	39.5%	26	740
Arbois	3,640	185	8.5	36.2%	34	850
Roques	3,210	310	17.4	19.8%	9	35
Sambuc	2,980	450	18.9	15.2%	7	20
Fontblanche	2,750	560	20.1	12.8%	6	15
Lançon	2,640	165	10.2	31.5%	16	180

Massif Name	Area (ha)	Mean Elev (m)	Mean Slope (°)	WUI Area (%)	Fires (2010–2025)	Burned (ha)
Sulauze	2,420	110	7.8	26.4%	11	60
La Fare	2,150	170	12.4	33.1%	14	95
Rougadou	1,940	160	11.8	27.5%	10	40
Saint-Ambroise	1,820	125	9.5	29.2%	8	30
Marignane	1,530	85	8.1	58.4%	17	210
Petit Luberon	1,480	490	23.5	11.4%	4	10
Tallagard	1,320	210	14.8	30.6%	7	25
Saint-Cannat	1,180	240	12.1	25.8%	5	15
Gardiole	980	180	21.6	37.2%	11	85